# CNNP Skill 3: Queueing Curve, Little's Law and Buffer Analysis

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## 1. Install tools and verify versions

```
sudo apt update
sudo apt install -y iperf3 iproute2 wireshark tshark iputils-ping
iperf3 --version
tc -V
```

**Observation:** iperf 3.20 (cJSON 1.7.15) and tc utility iproute2-6.19.0 are installed.

```
tejaswin@DESKTOP-P4G181S:~$ sudo apt update
sudo apt install -y iperf3 iproute2 wireshark tshark iputils-ping

# Verify installations
iperf3 --version
tc -V
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
139 packages can be upgraded. Run 'apt list --upgradable' to see them.
iperf3 is already the newest version (3.20-2).
iproute2 is already the newest version (6.19.0-1ubuntu1.1).
wireshark is already the newest version (4.6.4-1).
tshark is already the newest version (4.6.4-1).
iputils-ping is already the newest version (3:20250605-1ubuntu1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 139
iperf 3.20 (cJSON 1.7.15)
Linux DESKTOP-P4G181S 6.18.58.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Tue Oct 27 00:24:11 UTC 2026 x86_64
Optional features available: CPU affinity setting, IPv6 flow label, SCTP, TCP congestion algorithm setting, sendfile / zerocop
y, socket pacing, authentication, bind to device, support IPv4 don't fragment, POSIX threads
tc utility, iproute2-6.19.0, libbpf 1.6.3
```

*Figure: apt update and tool version checks*

## 2. Topology: sender, router and receiver

```
sudo ip netns add sender
sudo ip netns add router
sudo ip netns add receiver
sudo ip link add veth-s type veth peer name veth-r0
sudo ip link add veth-r1 type veth peer name veth-r
sudo ip netns exec router ip link set veth-r0 name eth0
sudo ip netns exec router ip link set veth-r1 name eth1
sudo ip netns exec sender ip addr add 192.168.10.2/24 dev veth-s
sudo ip netns exec router ip addr add 192.168.10.1/24 dev eth0
sudo ip netns exec router ip addr add 192.168.20.1/24 dev eth1
sudo ip netns exec receiver ip addr add 192.168.20.2/24 dev veth-r
sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec sender ip route add 192.168.20.0/24 via 192.168.10.1
sudo ip netns exec receiver ip route add 192.168.10.0/24 via 192.168.20.1
sudo ip netns exec sender ping -c 4 192.168.20.2
```

**Observation:** ICMP replies from 192.168.20.2 with TTL 63 (one router hop).

```
tejaswin@DESKTOP-P4G1815:~$ # 1. Reset namespaces (restores host eth0/eth1 to root namespace)
sudo ip netns del sender 2>/dev/null || true
sudo ip netns del router 2>/dev/null || true
sudo ip netns del receiver 2>/dev/null || true

# 2. Recreate namespaces and veth pairs with unique root-level names
sudo ip netns add sender
sudo ip netns add router
sudo ip netns add receiver

sudo ip link add veth-s type veth peer name veth-r0
sudo ip link add veth-r1 type veth peer name veth-r

sudo ip link set veth-s netns sender
sudo ip link set veth-r0 netns router
sudo ip link set veth-r1 netns router
sudo ip link set veth-r netns receiver

# 3. Rename interfaces inside router to eth0 and eth1
sudo ip netns exec router ip link set veth-r0 name eth0
sudo ip netns exec router ip link set veth-r1 name eth1

# 4. Configure Sender (192.168.10.2)
sudo ip netns exec sender ip addr add 192.168.10.2/24 dev veth-s
sudo ip netns exec sender ip link set veth-s up
sudo ip netns exec sender ip link set lo up

# 5. Configure Router (eth0 = 192.168.10.1, eth1 = 192.168.20.1)
sudo ip netns exec router ip addr add 192.168.10.1/24 dev eth0
sudo ip netns exec router ip addr add 192.168.20.1/24 dev eth1
sudo ip netns exec router ip link set eth0 up
sudo ip netns exec router ip link set eth1 up
sudo ip netns exec router ip link set lo up

# 6. Configure Receiver (192.168.20.2)
sudo ip netns exec receiver ip addr add 192.168.20.2/24 dev veth-r
sudo ip netns exec receiver ip link set veth-r up
sudo ip netns exec receiver ip link set lo up

# 7. Enable IPv4 forwarding and static routes
sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec router ip addr
sudo ip netns exec sender ip route add 192.168.20.0/24 via 192.168.10.1
sudo ip netns exec receiver ip route add 192.168.10.0/24 via 192.168.20.1

# 8. Test connectivity
sudo ip netns exec sender ping -c 4 192.168.20.2
net.ipv4.ip_forward = 1
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
       valid_lft forever preferred_lft forever
7: eth0@if8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 12:d5:e6:79:a6:de brd ff:ff:ff:ff:ff:ff link-netns sender
    inet 192.168.10.1/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::10d5:e6ff:fe79:a6de/64 scope link tentative proto kernel_ll
       valid_lft forever preferred_lft forever
10: eth1@if9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether e6:b1:09:fa:79:d5 brd ff:ff:ff:ff:ff:ff link-netns receiver
    inet 192.168.20.1/24 scope global eth1
       valid_lft forever preferred_lft forever
    inet6 fe80::e4b1:9ff:fefa:79d5/64 scope link tentative proto kernel_ll
       valid_lft forever preferred_lft forever
PING 192.168.20.2 (192.168.20.2) 56(84) bytes of data.
64 bytes from 192.168.20.2: icmp_seq=1 ttl=63 time=0.047 ms
64 bytes from 192.168.20.2: icmp_seq=2 ttl=63 time=0.072 ms
64 bytes from 192.168.20.2: icmp_seq=3 ttl=63 time=0.045 ms
64 bytes from 192.168.20.2: icmp_seq=4 ttl=63 time=0.054 ms

--- 192.168.20.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3052ms
rtt min/avg/max/mdev = 0.045/0.054/0.072/0.010 ms
```

*Figure: Namespace, address and routing setup, with connectivity ping*

## 3. Token bucket bottleneck on router eth1

```
sudo ip netns exec router tc qdisc replace dev eth1 root tbf rate 5mbit burst 32kb limit 64kb
sudo ip netns exec router tc -s qdisc show dev eth1
```

**Observation:** qdisc tbf 8001, rate 5Mbit, burst 32Kb, lat 52.4ms. Sent 0 bytes, 0 packets, dropped 0, overlimits 0, requeues 0 before traffic starts.

```
tejaswin@DESKTOP-P4G181S:~$ sudo ip netns exec router tc qdisc replace dev eth1 root tbf rate 5mbit burst 32kb limit 64kb
sudo ip netns exec router tc -s qdisc show dev eth1
qdisc tbf 8001: root refcnt 9 rate 5Mbit burst 32Kb lat 52.4ms
 Sent 0 bytes 0 pkt (dropped 0, overlimits 0 requeues 0)
 backlog 0b 0p requeues 0
```

*Figure: tc tbf configuration and statistics*

## 4. Receiver server and router capture

```
sudo ip netns exec receiver iperf3 -s
sudo ip netns exec router tshark -i eth1 -w /tmp/Queueing_Little_Law_Buffer_Analysis.pcapng
```

**Observation:** the receiver is listening on port 5201 and the router capture on eth1 is running.

```
tejaswin@DESKTOP-P4G181S:~$ sudo ip netns exec receiver iperf3 -s
[sudo: authenticate] Password:
-----------------------------------------------------------
Server listening on 5201 (test #1)
-----------------------------------------------------------
```

*Figure: iperf3 receiver listening on port 5201*

```
tejaswin@DESKTOP-P4G181S:~$ sudo ip netns exec router tshark -i eth1 -w /tmp/Queueing_Little_Law_Buffer_Analysis.pcapng
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth1'
```

*Figure: Router capture on eth1 started*

## 5. Queueing curve: UDP load tests

```
sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 2M -l 1200 -t 30 -i 1
sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 4M -l 1200 -t 30 -i 1
sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 5M -l 1200 -t 30 -i 1
sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 7M -l 1200 -t 30 -i 1
sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 10M -l 1200 -t 30 -i 1
```

```
tejaswin@DESKTOP-P4G1815:~$ # Test 1: 1 Mbps
sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 1M -l 1200 -t 30 -i 1

# Test 2: 2 Mbps
sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 2M -l 1200 -t 30 -i 1

# Test 3: 4 Mbps
sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 4M -l 1200 -t 30 -i 1

# Test 4: 5 Mbps (Knee / bottleneck threshold)
sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 5M -l 1200 -t 30 -i 1

# Test 5: 7 Mbps (Congested)
sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 7M -l 1200 -t 30 -i 1

# Test 6: 10 Mbps (Heavy congestion)
sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 10M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 44104 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   123 KBytes  1.01 Mbits/sec  105
[  5]   1.00-2.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   2.00-3.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   3.00-4.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   4.00-5.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   5.00-6.00   sec   123 KBytes  1.01 Mbits/sec  105
[  5]   6.00-7.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   7.00-8.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   8.00-9.00   sec   122 KBytes   999 Kbits/sec  104
[  5]   9.00-10.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  10.00-11.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  11.00-12.00  sec   122 KBytes   999 Kbits/sec  105
[  5]  12.00-13.00  sec   123 KBytes  1.01 Mbits/sec  104
[  5]  13.00-14.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  14.00-15.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  15.00-16.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  16.00-17.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  17.00-18.00  sec   122 KBytes   999 Kbits/sec  105
[  5]  18.00-19.00  sec   123 KBytes  1.01 Mbits/sec  104
[  5]  19.00-20.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  20.00-21.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  21.00-22.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  22.00-23.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  23.00-24.00  sec   122 KBytes   999 Kbits/sec  105
[  5]  24.00-25.00  sec   123 KBytes  1.01 Mbits/sec  104
[  5]  25.00-26.00  sec   122 KBytes   999 Kbits/sec  104
[  5]  26.00-27.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  27.00-28.00  sec   122 KBytes   999 Kbits/sec  104
[  5]  28.00-29.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  29.00-30.04  sec   123 KBytes   970 Kbits/sec  105
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.04  sec  3.58 MBytes   999 Kbits/sec  0.000 ms  0/3126 (0%)  sender
[  5]   0.00-30.04  sec  3.58 MBytes   999 Kbits/sec  0.114 ms  0/3126 (0%)  receiver
```

*Figure: Load test commands and labels*

```
iperf Done.
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 37548 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   245 KBytes  2.00 Mbits/sec  209
[  5]   1.00-2.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   2.00-3.00   sec   245 KBytes  2.01 Mbits/sec  209
[  5]   3.00-4.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   4.00-5.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   5.00-6.00   sec   245 KBytes  2.01 Mbits/sec  209
[  5]   6.00-7.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   7.00-8.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   8.00-9.00   sec   245 KBytes  2.01 Mbits/sec  209
[  5]   9.00-10.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  10.00-11.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  11.00-12.00  sec   245 KBytes  2.01 Mbits/sec  209
[  5]  12.00-13.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  13.00-14.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  14.00-15.00  sec   245 KBytes  2.01 Mbits/sec  209
[  5]  15.00-16.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  16.00-17.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  17.00-18.00  sec   245 KBytes  2.01 Mbits/sec  209
[  5]  18.00-19.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  19.00-20.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  20.00-21.00  sec   245 KBytes  2.01 Mbits/sec  209
[  5]  21.00-22.00  sec   244 KBytes  1.99 Mbits/sec  208
[  5]  22.00-23.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  23.00-24.00  sec   245 KBytes  2.01 Mbits/sec  209
[  5]  24.00-25.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  25.00-26.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  26.00-27.00  sec   245 KBytes  2.01 Mbits/sec  209
[  5]  27.00-28.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  28.00-29.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  29.00-30.00  sec   245 KBytes  2.00 Mbits/sec  209
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  7.15 MBytes  2.00 Mbits/sec  0.000 ms  0/6251 (0%)  sender
[  5]   0.00-30.00  sec  7.15 MBytes  2.00 Mbits/sec  0.050 ms  0/6251 (0%)  receiver
```

*Figure: Test 1: 2 Mbit/s*

```
iperf Done.
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 50691 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   490 KBytes  4.01 Mbits/sec  418
[  5]   1.00-2.00   sec   488 KBytes  3.99 Mbits/sec  416
[  5]   2.00-3.00   sec   489 KBytes  4.00 Mbits/sec  417
[  5]   3.00-4.00   sec   489 KBytes  4.00 Mbits/sec  417
[  5]   4.00-5.00   sec   488 KBytes  3.99 Mbits/sec  416
[  5]   5.00-6.00   sec   489 KBytes  4.00 Mbits/sec  417
[  5]   6.00-7.00   sec   488 KBytes  3.99 Mbits/sec  416
[  5]   7.00-8.00   sec   489 KBytes  4.00 Mbits/sec  417
[  5]   8.00-9.00   sec   489 KBytes  4.00 Mbits/sec  417
[  5]   9.00-10.00  sec   488 KBytes  3.99 Mbits/sec  417
[  5]  10.00-11.00  sec   489 KBytes  4.00 Mbits/sec  416
[  5]  11.00-12.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  12.00-13.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  13.00-14.00  sec   488 KBytes  3.99 Mbits/sec  416
[  5]  14.00-15.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  15.00-16.00  sec   488 KBytes  3.99 Mbits/sec  417
[  5]  16.00-17.00  sec   489 KBytes  4.00 Mbits/sec  416
[  5]  17.00-18.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  18.00-19.00  sec   488 KBytes  3.99 Mbits/sec  416
[  5]  19.00-20.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  20.00-21.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  21.00-22.00  sec   488 KBytes  3.99 Mbits/sec  416
[  5]  22.00-23.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  23.00-24.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  24.00-25.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  25.00-26.00  sec   488 KBytes  3.99 Mbits/sec  416
[  5]  26.00-27.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  27.00-28.00  sec   488 KBytes  3.99 Mbits/sec  417
[  5]  28.00-29.00  sec   489 KBytes  4.00 Mbits/sec  416
[  5]  29.00-30.00  sec   489 KBytes  4.00 Mbits/sec  417
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  14.3 MBytes  4.00 Mbits/sec  0.000 ms  0/12501 (0%)  sender
[  5]   0.00-30.00  sec  14.3 MBytes  4.00 Mbits/sec  0.055 ms  0/12501 (0%)  receiver
```

*Figure: Test 2: 4 Mbit/s*

```
iperf Done.
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 50909 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   612 KBytes  5.01 Mbits/sec  522
[  5]   1.00-2.00   sec   609 KBytes  4.99 Mbits/sec  520
[  5]   2.00-3.00   sec   612 KBytes  5.01 Mbits/sec  522
[  5]   3.00-4.00   sec   609 KBytes  4.99 Mbits/sec  520
[  5]   4.00-5.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   5.00-6.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   6.00-7.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   7.00-8.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   8.00-9.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   9.00-10.00  sec   609 KBytes  4.99 Mbits/sec  520
[  5]  10.00-11.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  11.00-12.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  12.00-13.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  13.00-14.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  14.00-15.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  15.00-16.00  sec   609 KBytes  4.99 Mbits/sec  520
[  5]  16.00-17.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  17.00-18.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  18.00-19.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  19.00-20.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  20.00-21.00  sec   609 KBytes  4.99 Mbits/sec  520
[  5]  21.00-22.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  22.00-23.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  23.00-24.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  24.00-25.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  25.00-26.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  26.00-27.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  27.00-28.00  sec   609 KBytes  4.99 Mbits/sec  520
[  5]  28.00-29.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  29.00-30.00  sec   611 KBytes  5.00 Mbits/sec  521
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  17.9 MBytes  5.00 Mbits/sec  0.000 ms  0/15626 (0%)  sender
[  5]   0.00-30.10  sec  17.4 MBytes  4.84 Mbits/sec  0.284 ms  452/15626 (2.9%)  receiver
```

*Figure: Test 3: 5 Mbit/s (bottleneck)*

```
iperf Done.
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 55680 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   855 KBytes  7.00 Mbits/sec  730
[  5]   1.00-2.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   2.00-3.00   sec   855 KBytes  7.01 Mbits/sec  730
[  5]   3.00-4.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   4.00-5.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   5.00-6.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   6.00-7.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   7.00-8.00   sec   855 KBytes  7.01 Mbits/sec  730
[  5]   8.00-9.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   9.00-10.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  10.00-11.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  11.00-12.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  12.00-13.00  sec   854 KBytes  6.99 Mbits/sec  729
[  5]  13.00-14.00  sec   855 KBytes  7.01 Mbits/sec  730
[  5]  14.00-15.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  15.00-16.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  16.00-17.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  17.00-18.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  18.00-19.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  19.00-20.00  sec   854 KBytes  7.00 Mbits/sec  730
[  5]  20.00-21.00  sec   855 KBytes  7.01 Mbits/sec  729
[  5]  21.00-22.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  22.00-23.00  sec   854 KBytes  6.99 Mbits/sec  729
[  5]  23.00-24.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  24.00-25.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  25.00-26.00  sec   854 KBytes  7.00 Mbits/sec  730
[  5]  26.00-27.00  sec   855 KBytes  7.01 Mbits/sec  729
[  5]  27.00-28.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  28.00-29.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  29.00-30.00  sec   854 KBytes  7.00 Mbits/sec  729
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  25.0 MBytes  7.00 Mbits/sec  0.000 ms  0/21876 (0%)  sender
[  5]   0.00-30.10  sec  17.3 MBytes  4.83 Mbits/sec  0.721 ms  6715/21875 (31%)  receiver
```

*Figure: Test 4: 7 Mbit/s (congested)*

```
iperf Done.
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 36702 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec  1.19 MBytes  10.0 Mbits/sec  1043
[  5]   1.00-2.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]   2.00-3.00   sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]   3.00-4.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]   4.00-5.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]   5.00-6.00   sec  1.19 MBytes  9.99 Mbits/sec  1042
[  5]   6.00-7.00   sec  1.19 MBytes  10.0 Mbits/sec  1041
[  5]   7.00-8.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]   8.00-9.00   sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]   9.00-10.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  10.00-11.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  11.00-12.00  sec  1.19 MBytes  9.99 Mbits/sec  1042
[  5]  12.00-13.00  sec  1.19 MBytes  10.0 Mbits/sec  1041
[  5]  13.00-14.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  14.00-15.00  sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]  15.00-16.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  16.00-17.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  17.00-18.00  sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]  18.00-19.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  19.00-20.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  20.00-21.00  sec  1.19 MBytes  9.99 Mbits/sec  1042
[  5]  21.00-22.00  sec  1.19 MBytes  10.0 Mbits/sec  1041
[  5]  22.00-23.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  23.00-24.00  sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]  24.00-25.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  25.00-26.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  26.00-27.00  sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]  27.00-28.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  28.00-29.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  29.00-30.00  sec  1.19 MBytes  9.99 Mbits/sec  1041
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  35.8 MBytes  10.0 Mbits/sec  0.000 ms  0/31251 (0%)  sender
[  5]   0.00-30.10  sec  17.4 MBytes  4.83 Mbits/sec  0.213 ms  16090/31251 (51%)  receiver

iperf Done.
tejaswin@DESKTOP-P4G181S:~$ sudo chmod 644 /tmp/Queueing_Little_Law_Buffer_Analysis.pcapng
cp /tmp/Queueing_Little_Law_Buffer_Analysis.pcapng ~/
```

*Figure: Test 5: 10 Mbit/s (heavy congestion)*

| Offered rate | Sender bitrate | Transfer (30 s) | Receiver bitrate | Receiver loss | Receiver jitter |
|---|---|---|---|---|---|
| 2 Mbit/s | 2.00 Mbit/s | 7.15 MB | 2.00 Mbit/s | 0 / 6251 (0%) | 0.050 ms |
| 4 Mbit/s | 4.00 Mbit/s | 14.3 MB | 4.00 Mbit/s | 0 / 12501 (0%) | 0.055 ms |
| 5 Mbit/s (bottleneck) | 5.00 Mbit/s | 17.9 MB | 4.84 Mbit/s | 452 / 15626 (2.9%) | 0.284 ms |
| 7 Mbit/s (congested) | 7.00 Mbit/s | 25.0 MB | 4.83 Mbit/s | 6715 / 21875 (31%) | 0.721 ms |
| 10 Mbit/s (heavy) | 10.0 Mbit/s | 35.8 MB | 4.83 Mbit/s | 16090 / 31251 (51%) | 0.213 ms |

**Observation:** below 5 Mbit/s the receiver got every datagram. At 5 Mbit/s the token bucket starts dropping (2.9% loss). Above that, receiver throughput stays near 4.83 Mbit/s while loss rises to 31% and 51%, consistent with queue overflow at the eth1 limit.

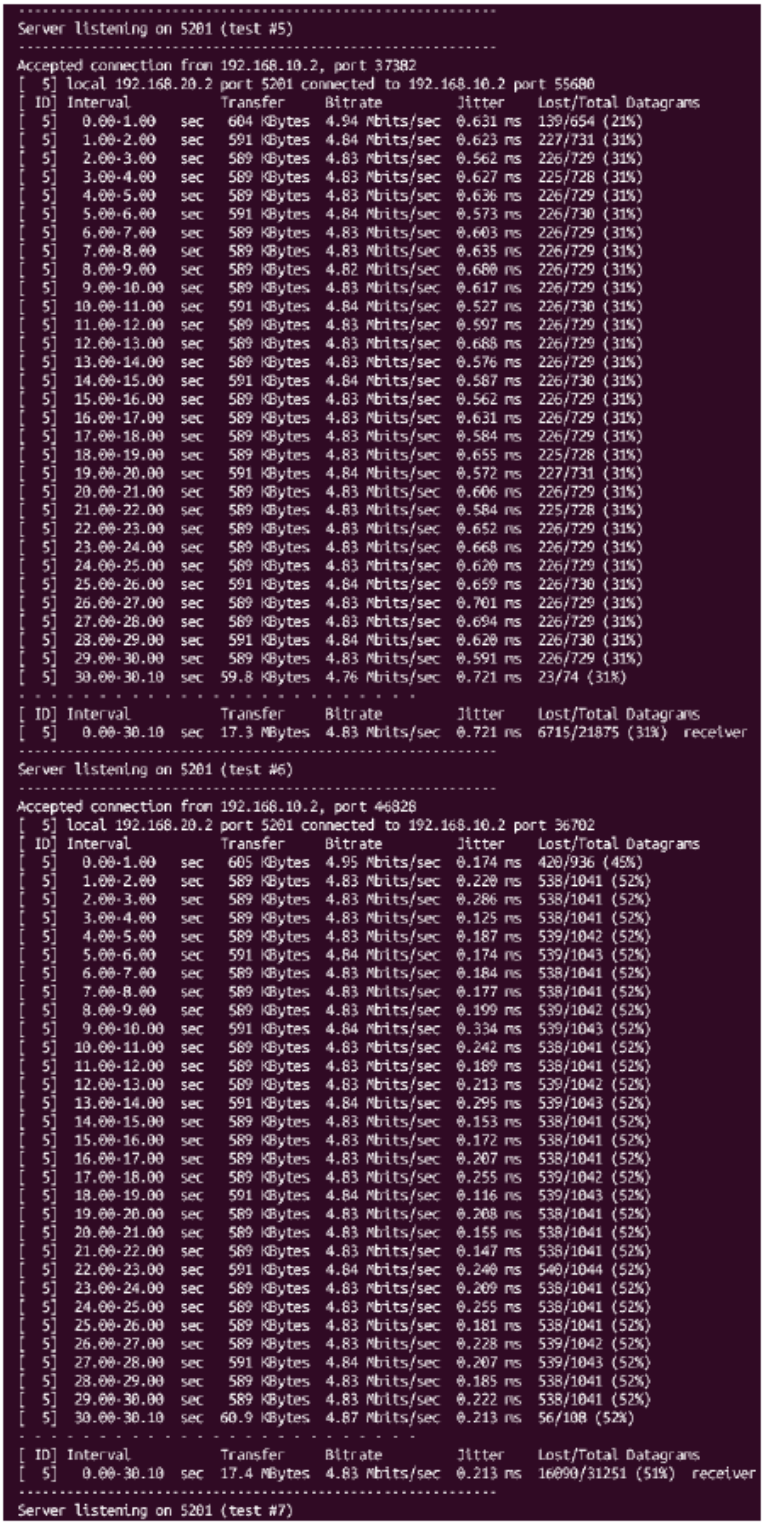
## 6. Receiver logs for the remaining test runs

```
tejaswin@DESKTOP-P4G1B1S:~$ sudo ip netns exec receiver iperf3 -s
[sudo: authenticate] Password:
-----------------------------------------------------------
Server listening on 5201 (test #1)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 53362
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 44104
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   123 KBytes  1.01 Mbits/sec  0.031 ms  0/105 (0%)
[  5]   1.00-2.00   sec   122 KBytes   998 Kbits/sec  0.045 ms  0/104 (0%)
[  5]   2.00-3.00   sec   122 KBytes   998 Kbits/sec  0.041 ms  0/104 (0%)
[  5]   3.00-4.00   sec   122 KBytes   998 Kbits/sec  0.049 ms  0/104 (0%)
[  5]   4.00-5.00   sec   122 KBytes   998 Kbits/sec  0.054 ms  0/104 (0%)
[  5]   5.00-6.00   sec   123 KBytes  1.01 Mbits/sec  0.065 ms  0/105 (0%)
[  5]   6.00-7.00   sec   122 KBytes   998 Kbits/sec  0.050 ms  0/104 (0%)
[  5]   7.00-8.00   sec   122 KBytes   998 Kbits/sec  0.128 ms  0/104 (0%)
[  5]   8.00-9.00   sec   122 KBytes   998 Kbits/sec  0.057 ms  0/104 (0%)
[  5]   9.00-10.00  sec   122 KBytes   999 Kbits/sec  0.041 ms  0/104 (0%)
[  5]  10.00-11.00  sec   122 KBytes   998 Kbits/sec  0.030 ms  0/104 (0%)
[  5]  11.00-12.00  sec   123 KBytes  1.01 Mbits/sec  0.027 ms  0/105 (0%)
[  5]  12.00-13.00  sec   122 KBytes   998 Kbits/sec  0.062 ms  0/104 (0%)
[  5]  13.00-14.00  sec   122 KBytes   998 Kbits/sec  0.087 ms  0/104 (0%)
[  5]  14.00-15.00  sec   122 KBytes   998 Kbits/sec  0.056 ms  0/104 (0%)
[  5]  15.00-16.00  sec   122 KBytes   998 Kbits/sec  0.042 ms  0/104 (0%)
[  5]  16.00-17.00  sec   122 KBytes   998 Kbits/sec  0.073 ms  0/104 (0%)
[  5]  17.00-18.00  sec   123 KBytes  1.01 Mbits/sec  0.082 ms  0/105 (0%)
[  5]  18.00-19.00  sec   122 KBytes   998 Kbits/sec  0.066 ms  0/104 (0%)
[  5]  19.00-20.00  sec   122 KBytes   998 Kbits/sec  0.056 ms  0/104 (0%)
[  5]  20.00-21.00  sec   122 KBytes   998 Kbits/sec  0.078 ms  0/104 (0%)
[  5]  21.00-22.00  sec   122 KBytes   998 Kbits/sec  0.087 ms  0/104 (0%)
[  5]  22.00-23.00  sec   122 KBytes   998 Kbits/sec  0.062 ms  0/104 (0%)
[  5]  23.00-24.00  sec   123 KBytes  1.01 Mbits/sec  0.043 ms  0/105 (0%)
[  5]  24.00-25.00  sec   122 KBytes   998 Kbits/sec  0.078 ms  0/104 (0%)
[  5]  25.00-26.00  sec   122 KBytes   999 Kbits/sec  0.020 ms  0/104 (0%)
[  5]  26.00-27.00  sec   122 KBytes   998 Kbits/sec  0.057 ms  0/104 (0%)
[  5]  27.00-28.00  sec   122 KBytes   999 Kbits/sec  0.077 ms  0/104 (0%)
[  5]  28.00-29.00  sec   122 KBytes   998 Kbits/sec  0.044 ms  0/104 (0%)
[  5]  29.00-30.00  sec   123 KBytes  1.01 Mbits/sec  0.114 ms  0/105 (0%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.04  sec  3.58 MBytes   999 Kbits/sec  0.114 ms  0/3126 (0%)  receiver
-----------------------------------------------------------
Server listening on 5201 (test #2)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 38030
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 37548
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   245 KBytes  2.00 Mbits/sec  0.054 ms  0/209 (0%)
[  5]   1.00-2.00   sec   244 KBytes  2.00 Mbits/sec  0.050 ms  0/208 (0%)
[  5]   2.00-3.00   sec   245 KBytes  2.01 Mbits/sec  0.067 ms  0/209 (0%)
[  5]   3.00-4.00   sec   244 KBytes  2.00 Mbits/sec  0.030 ms  0/208 (0%)
[  5]   4.00-5.00   sec   244 KBytes  2.00 Mbits/sec  0.052 ms  0/208 (0%)
[  5]   5.00-6.00   sec   245 KBytes  2.01 Mbits/sec  0.108 ms  0/209 (0%)
[  5]   6.00-7.00   sec   244 KBytes  2.00 Mbits/sec  0.038 ms  0/208 (0%)
[  5]   7.00-8.00   sec   244 KBytes  2.00 Mbits/sec  0.075 ms  0/208 (0%)
[  5]   8.00-9.00   sec   245 KBytes  2.01 Mbits/sec  0.067 ms  0/209 (0%)
[  5]   9.00-10.00  sec   244 KBytes  2.00 Mbits/sec  0.079 ms  0/208 (0%)
[  5]  10.00-11.00  sec   244 KBytes  2.00 Mbits/sec  0.063 ms  0/208 (0%)
[  5]  11.00-12.00  sec   245 KBytes  2.01 Mbits/sec  0.099 ms  0/209 (0%)
[  5]  12.00-13.00  sec   244 KBytes  2.00 Mbits/sec  0.059 ms  0/208 (0%)
[  5]  13.00-14.00  sec   244 KBytes  2.00 Mbits/sec  0.050 ms  0/208 (0%)
[  5]  14.00-15.00  sec   245 KBytes  2.01 Mbits/sec  0.154 ms  0/209 (0%)
[  5]  15.00-16.00  sec   244 KBytes  2.00 Mbits/sec  0.056 ms  0/208 (0%)
[  5]  16.00-17.00  sec   244 KBytes  2.00 Mbits/sec  0.058 ms  0/208 (0%)
[  5]  17.00-18.00  sec   245 KBytes  2.01 Mbits/sec  0.073 ms  0/209 (0%)
[  5]  18.00-19.00  sec   244 KBytes  2.00 Mbits/sec  0.045 ms  0/208 (0%)
[  5]  19.00-20.00  sec   244 KBytes  2.00 Mbits/sec  0.054 ms  0/208 (0%)
[  5]  20.00-21.00  sec   245 KBytes  2.01 Mbits/sec  0.045 ms  0/209 (0%)
[  5]  21.00-22.00  sec   244 KBytes  2.00 Mbits/sec  0.052 ms  0/208 (0%)
[  5]  22.00-23.00  sec   244 KBytes  2.00 Mbits/sec  0.039 ms  0/208 (0%)
[  5]  23.00-24.00  sec   245 KBytes  2.01 Mbits/sec  0.038 ms  0/209 (0%)
[  5]  24.00-25.00  sec   244 KBytes  2.00 Mbits/sec  0.053 ms  0/208 (0%)
[  5]  25.00-26.00  sec   244 KBytes  2.00 Mbits/sec  0.083 ms  0/208 (0%)
[  5]  26.00-27.00  sec   245 KBytes  2.01 Mbits/sec  0.047 ms  0/209 (0%)
[  5]  27.00-28.00  sec   244 KBytes  2.00 Mbits/sec  0.090 ms  0/208 (0%)
[  5]  28.00-29.00  sec   244 KBytes  2.00 Mbits/sec  0.043 ms  0/208 (0%)
[  5]  29.00-30.00  sec   245 KBytes  2.01 Mbits/sec  0.050 ms  0/209 (0%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  7.15 MBytes  2.00 Mbits/sec  0.050 ms  0/6251 (0%)  receiver
```

*Figure: Receiver server log (multiple runs)*

```
-----------------------------------------------------------
Server listening on 5201 (test #3)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 43958
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 50691
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   489 KBytes  4.00 Mbits/sec  0.047 ms  0/417 (0%)
[  5]   1.00-2.00   sec   489 KBytes  4.00 Mbits/sec  0.047 ms  0/417 (0%)
[  5]   2.00-3.00   sec   489 KBytes  4.00 Mbits/sec  0.073 ms  0/417 (0%)
[  5]   3.00-4.00   sec   489 KBytes  4.00 Mbits/sec  0.053 ms  0/417 (0%)
[  5]   4.00-5.00   sec   488 KBytes  3.99 Mbits/sec  0.051 ms  0/416 (0%)
[  5]   5.00-6.00   sec   489 KBytes  4.00 Mbits/sec  0.069 ms  0/417 (0%)
[  5]   6.00-7.00   sec   488 KBytes  3.99 Mbits/sec  0.085 ms  0/416 (0%)
[  5]   7.00-8.00   sec   489 KBytes  4.00 Mbits/sec  0.054 ms  0/417 (0%)
[  5]   8.00-9.00   sec   489 KBytes  4.00 Mbits/sec  0.067 ms  0/417 (0%)
[  5]   9.00-10.00  sec   488 KBytes  3.99 Mbits/sec  0.072 ms  0/416 (0%)
[  5]  10.00-11.00  sec   489 KBytes  4.00 Mbits/sec  0.064 ms  0/417 (0%)
[  5]  11.00-12.00  sec   489 KBytes  4.00 Mbits/sec  0.079 ms  0/417 (0%)
[  5]  12.00-13.00  sec   488 KBytes  3.99 Mbits/sec  0.064 ms  0/416 (0%)
[  5]  13.00-14.00  sec   489 KBytes  4.00 Mbits/sec  0.054 ms  0/417 (0%)
[  5]  14.00-15.00  sec   489 KBytes  4.00 Mbits/sec  0.107 ms  0/417 (0%)
[  5]  15.00-16.00  sec   488 KBytes  3.99 Mbits/sec  0.058 ms  0/416 (0%)
[  5]  16.00-17.00  sec   489 KBytes  4.00 Mbits/sec  0.046 ms  0/417 (0%)
[  5]  17.00-18.00  sec   489 KBytes  4.00 Mbits/sec  0.064 ms  0/417 (0%)
[  5]  18.00-19.00  sec   488 KBytes  3.99 Mbits/sec  0.042 ms  0/416 (0%)
[  5]  19.00-20.00  sec   489 KBytes  4.00 Mbits/sec  0.048 ms  0/417 (0%)
[  5]  20.00-21.00  sec   489 KBytes  4.00 Mbits/sec  0.062 ms  0/417 (0%)
[  5]  21.00-22.00  sec   488 KBytes  3.99 Mbits/sec  0.038 ms  0/416 (0%)
[  5]  22.00-23.00  sec   489 KBytes  4.00 Mbits/sec  0.131 ms  0/417 (0%)
[  5]  23.00-24.00  sec   489 KBytes  4.00 Mbits/sec  0.146 ms  0/417 (0%)
[  5]  24.00-25.00  sec   489 KBytes  4.00 Mbits/sec  0.041 ms  0/417 (0%)
[  5]  25.00-26.00  sec   488 KBytes  3.99 Mbits/sec  0.055 ms  0/416 (0%)
[  5]  26.00-27.00  sec   489 KBytes  4.00 Mbits/sec  0.063 ms  0/417 (0%)
[  5]  27.00-28.00  sec   488 KBytes  3.99 Mbits/sec  0.068 ms  0/416 (0%)
[  5]  28.00-29.00  sec   489 KBytes  4.00 Mbits/sec  0.042 ms  0/417 (0%)
[  5]  29.00-30.00  sec   489 KBytes  4.00 Mbits/sec  0.055 ms  0/417 (0%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  14.3 MBytes  4.00 Mbits/sec  0.055 ms  0/12501 (0%)  receiver
-----------------------------------------------------------
Server listening on 5201 (test #4)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 34120
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 50909
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   609 KBytes  4.99 Mbits/sec  0.070 ms  0/520 (0%)
[  5]   1.00-2.00   sec   600 KBytes  4.91 Mbits/sec  0.212 ms  0/512 (0%)
[  5]   2.00-3.00   sec   589 KBytes  4.83 Mbits/sec  0.139 ms  0/503 (0%)
[  5]   3.00-4.00   sec   589 KBytes  4.83 Mbits/sec  0.137 ms  0/503 (0%)
[  5]   4.00-5.00   sec   589 KBytes  4.83 Mbits/sec  0.145 ms  9/512 (1.8%)
[  5]   5.00-6.00   sec   591 KBytes  4.84 Mbits/sec  0.205 ms  18/522 (3.4%)
[  5]   6.00-7.00   sec   589 KBytes  4.83 Mbits/sec  0.202 ms  18/521 (3.5%)
[  5]   7.00-8.00   sec   589 KBytes  4.82 Mbits/sec  0.167 ms  17/520 (3.3%)
[  5]   8.00-9.00   sec   589 KBytes  4.83 Mbits/sec  0.190 ms  18/521 (3.5%)
[  5]   9.00-10.00  sec   591 KBytes  4.84 Mbits/sec  0.172 ms  17/521 (3.3%)
[  5]  10.00-11.00  sec   588 KBytes  4.82 Mbits/sec  0.147 ms  18/520 (3.5%)
[  5]  11.00-12.00  sec   591 KBytes  4.84 Mbits/sec  0.236 ms  18/522 (3.4%)
[  5]  12.00-13.00  sec   589 KBytes  4.82 Mbits/sec  0.152 ms  17/520 (3.3%)
[  5]  13.00-14.00  sec   589 KBytes  4.83 Mbits/sec  0.182 ms  18/521 (3.5%)
[  5]  14.00-15.00  sec   591 KBytes  4.84 Mbits/sec  0.220 ms  18/522 (3.4%)
[  5]  15.00-16.00  sec   589 KBytes  4.83 Mbits/sec  0.165 ms  17/520 (3.3%)
[  5]  16.00-17.00  sec   589 KBytes  4.83 Mbits/sec  0.252 ms  18/521 (3.5%)
[  5]  17.00-18.00  sec   591 KBytes  4.84 Mbits/sec  0.204 ms  17/521 (3.3%)
[  5]  18.00-19.00  sec   588 KBytes  4.82 Mbits/sec  0.172 ms  18/520 (3.5%)
[  5]  19.00-20.00  sec   591 KBytes  4.84 Mbits/sec  0.268 ms  18/522 (3.4%)
[  5]  20.00-21.00  sec   589 KBytes  4.83 Mbits/sec  0.153 ms  17/520 (3.3%)
[  5]  21.00-22.00  sec   589 KBytes  4.83 Mbits/sec  0.249 ms  18/521 (3.5%)
[  5]  22.00-23.00  sec   589 KBytes  4.83 Mbits/sec  0.149 ms  17/520 (3.3%)
[  5]  23.00-24.00  sec   589 KBytes  4.83 Mbits/sec  0.192 ms  18/521 (3.5%)
[  5]  24.00-25.00  sec   591 KBytes  4.84 Mbits/sec  0.235 ms  18/522 (3.4%)
[  5]  25.00-26.00  sec   589 KBytes  4.82 Mbits/sec  0.196 ms  17/520 (3.3%)
[  5]  26.00-27.00  sec   589 KBytes  4.83 Mbits/sec  0.202 ms  18/521 (3.5%)
[  5]  27.00-28.00  sec   589 KBytes  4.83 Mbits/sec  0.201 ms  17/520 (3.3%)
[  5]  28.00-29.00  sec   591 KBytes  4.84 Mbits/sec  0.161 ms  18/522 (3.4%)
[  5]  29.00-30.00  sec   589 KBytes  4.83 Mbits/sec  0.247 ms  18/521 (3.5%)
[  5]  30.00-30.10  sec  60.9 KBytes  4.89 Mbits/sec  0.284 ms  2/54 (3.7%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.10  sec  17.4 MBytes  4.84 Mbits/sec  0.284 ms  452/15626 (2.9%)  receiver
-----------------------------------------------------------
```

*Figure: Receiver server log (multiple runs)*



```
-----------------------------------------------------------
Server listening on 5201 (test #5)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 37382
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 55680
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   604 KBytes  4.94 Mbits/sec  0.631 ms  139/654 (21%)
[  5]   1.00-2.00   sec   591 KBytes  4.84 Mbits/sec  0.623 ms  227/731 (31%)
[  5]   2.00-3.00   sec   589 KBytes  4.83 Mbits/sec  0.562 ms  226/729 (31%)
[  5]   3.00-4.00   sec   589 KBytes  4.83 Mbits/sec  0.627 ms  225/728 (31%)
[  5]   4.00-5.00   sec   589 KBytes  4.83 Mbits/sec  0.636 ms  226/729 (31%)
[  5]   5.00-6.00   sec   591 KBytes  4.84 Mbits/sec  0.573 ms  226/730 (31%)
[  5]   6.00-7.00   sec   589 KBytes  4.83 Mbits/sec  0.603 ms  226/729 (31%)
[  5]   7.00-8.00   sec   589 KBytes  4.83 Mbits/sec  0.635 ms  226/729 (31%)
[  5]   8.00-9.00   sec   589 KBytes  4.82 Mbits/sec  0.680 ms  226/729 (31%)
[  5]   9.00-10.00  sec   589 KBytes  4.83 Mbits/sec  0.617 ms  226/729 (31%)
[  5]  10.00-11.00  sec   591 KBytes  4.84 Mbits/sec  0.527 ms  226/730 (31%)
[  5]  11.00-12.00  sec   589 KBytes  4.83 Mbits/sec  0.597 ms  226/729 (31%)
[  5]  12.00-13.00  sec   589 KBytes  4.83 Mbits/sec  0.688 ms  226/729 (31%)
[  5]  13.00-14.00  sec   589 KBytes  4.83 Mbits/sec  0.576 ms  226/729 (31%)
[  5]  14.00-15.00  sec   591 KBytes  4.84 Mbits/sec  0.587 ms  226/730 (31%)
[  5]  15.00-16.00  sec   589 KBytes  4.83 Mbits/sec  0.562 ms  226/729 (31%)
[  5]  16.00-17.00  sec   589 KBytes  4.83 Mbits/sec  0.631 ms  226/729 (31%)
[  5]  17.00-18.00  sec   589 KBytes  4.83 Mbits/sec  0.584 ms  226/729 (31%)
[  5]  18.00-19.00  sec   589 KBytes  4.83 Mbits/sec  0.655 ms  225/728 (31%)
[  5]  19.00-20.00  sec   591 KBytes  4.84 Mbits/sec  0.572 ms  227/731 (31%)
[  5]  20.00-21.00  sec   589 KBytes  4.83 Mbits/sec  0.606 ms  226/729 (31%)
[  5]  21.00-22.00  sec   589 KBytes  4.83 Mbits/sec  0.584 ms  225/728 (31%)
[  5]  22.00-23.00  sec   589 KBytes  4.83 Mbits/sec  0.652 ms  226/729 (31%)
[  5]  23.00-24.00  sec   589 KBytes  4.83 Mbits/sec  0.668 ms  226/729 (31%)
[  5]  24.00-25.00  sec   589 KBytes  4.83 Mbits/sec  0.620 ms  226/729 (31%)
[  5]  25.00-26.00  sec   591 KBytes  4.84 Mbits/sec  0.659 ms  226/730 (31%)
[  5]  26.00-27.00  sec   589 KBytes  4.83 Mbits/sec  0.701 ms  226/729 (31%)
[  5]  27.00-28.00  sec   589 KBytes  4.83 Mbits/sec  0.694 ms  226/729 (31%)
[  5]  28.00-29.00  sec   591 KBytes  4.84 Mbits/sec  0.620 ms  226/730 (31%)
[  5]  29.00-30.00  sec   589 KBytes  4.83 Mbits/sec  0.591 ms  226/729 (31%)
[  5]  30.00-30.10  sec  59.8 KBytes  4.76 Mbits/sec  0.721 ms  23/74 (31%)
- - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.10  sec  17.3 MBytes  4.83 Mbits/sec  0.721 ms  6715/21875 (31%)  receiver
-----------------------------------------------------------
Server listening on 5201 (test #6)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 46828
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 36702
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   605 KBytes  4.95 Mbits/sec  0.174 ms  420/936 (45%)
[  5]   1.00-2.00   sec   589 KBytes  4.83 Mbits/sec  0.220 ms  538/1041 (52%)
[  5]   2.00-3.00   sec   589 KBytes  4.83 Mbits/sec  0.286 ms  538/1041 (52%)
[  5]   3.00-4.00   sec   589 KBytes  4.83 Mbits/sec  0.125 ms  538/1041 (52%)
[  5]   4.00-5.00   sec   589 KBytes  4.83 Mbits/sec  0.187 ms  539/1042 (52%)
[  5]   5.00-6.00   sec   591 KBytes  4.84 Mbits/sec  0.174 ms  539/1043 (52%)
[  5]   6.00-7.00   sec   589 KBytes  4.83 Mbits/sec  0.184 ms  538/1041 (52%)
[  5]   7.00-8.00   sec   589 KBytes  4.83 Mbits/sec  0.177 ms  538/1041 (52%)
[  5]   8.00-9.00   sec   589 KBytes  4.83 Mbits/sec  0.199 ms  539/1042 (52%)
[  5]   9.00-10.00  sec   591 KBytes  4.84 Mbits/sec  0.334 ms  539/1043 (52%)
[  5]  10.00-11.00  sec   589 KBytes  4.83 Mbits/sec  0.242 ms  538/1041 (52%)
[  5]  11.00-12.00  sec   589 KBytes  4.83 Mbits/sec  0.189 ms  538/1041 (52%)
[  5]  12.00-13.00  sec   589 KBytes  4.83 Mbits/sec  0.213 ms  539/1042 (52%)
[  5]  13.00-14.00  sec   591 KBytes  4.84 Mbits/sec  0.295 ms  539/1043 (52%)
[  5]  14.00-15.00  sec   589 KBytes  4.83 Mbits/sec  0.153 ms  538/1041 (52%)
[  5]  15.00-16.00  sec   589 KBytes  4.83 Mbits/sec  0.172 ms  538/1041 (52%)
[  5]  16.00-17.00  sec   589 KBytes  4.83 Mbits/sec  0.207 ms  538/1041 (52%)
[  5]  17.00-18.00  sec   589 KBytes  4.83 Mbits/sec  0.255 ms  539/1042 (52%)
[  5]  18.00-19.00  sec   591 KBytes  4.84 Mbits/sec  0.116 ms  539/1043 (52%)
[  5]  19.00-20.00  sec   589 KBytes  4.83 Mbits/sec  0.208 ms  538/1041 (52%)
[  5]  20.00-21.00  sec   589 KBytes  4.83 Mbits/sec  0.155 ms  538/1041 (52%)
[  5]  21.00-22.00  sec   589 KBytes  4.83 Mbits/sec  0.147 ms  538/1041 (52%)
[  5]  22.00-23.00  sec   591 KBytes  4.84 Mbits/sec  0.240 ms  540/1044 (52%)
[  5]  23.00-24.00  sec   589 KBytes  4.83 Mbits/sec  0.209 ms  538/1041 (52%)
[  5]  24.00-25.00  sec   589 KBytes  4.83 Mbits/sec  0.255 ms  538/1041 (52%)
[  5]  25.00-26.00  sec   589 KBytes  4.83 Mbits/sec  0.181 ms  538/1041 (52%)
[  5]  26.00-27.00  sec   589 KBytes  4.83 Mbits/sec  0.228 ms  539/1042 (52%)
[  5]  27.00-28.00  sec   591 KBytes  4.84 Mbits/sec  0.207 ms  539/1043 (52%)
[  5]  28.00-29.00  sec   589 KBytes  4.83 Mbits/sec  0.185 ms  538/1041 (52%)
[  5]  29.00-30.00  sec   589 KBytes  4.83 Mbits/sec  0.222 ms  538/1041 (52%)
[  5]  30.00-30.10  sec  60.9 KBytes  4.87 Mbits/sec  0.213 ms  56/108 (52%)
- - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.10  sec  17.4 MBytes  4.83 Mbits/sec  0.213 ms  16090/31251 (51%)  receiver
-----------------------------------------------------------
Server listening on 5201 (test #7)
```

*Figure: Receiver server log (multiple runs)*

```
tejaswin@DESKTOP-P4G181S:~$ sudo ip netns exec router tshark -i eth1 -w /tmp/Queueing_Little_Law_Buffer_Analysis.pcapng
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth1'
67571
```

*Figure: Second router capture on eth1 (67571 packets counter)*

## 7. Little's Law (L = lambda x W) and buffer analysis

**Observation:** PENDING. From Test 5, receiver goodput is about 4.83 Mbit/s, or roughly 503 datagrams/s at 1200 bytes. L (backlog) and W (per-packet delay) need the tc backlog lines and loaded RTT, which are not verified. Buffer-limit results (16 kb, 64 kb, 256 kb) are not verified.

## 8. Cleanup

```
sudo ip netns del sender
sudo ip netns del router
sudo ip netns del receiver
sudo ip netns list
```